Avinash Dasari

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Education

Sandip University Nashik, India
B.Tech in Computer Science & Engineering (Specialization in AI & ML) 2022 - 2026
CGPA: 8.1
Relevant Coursework: OOP, Machine Learning, Deep Learning, Computer Vision, Image Processing, Internet of Things (IoT)

Technical Skills

Programming Languages	Python
AI/ML & Computer Vision	PyTorch, TensorFlow, OpenCV, Ultralytics YOLO, Scikit-learn, Image Processing
Data Analysis & Visualization	NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly
Application & Interface Tools	Qt5/6, Streamlit , pyinstaller
Databases	SQL, MySQL, MongoDB
Core Computer Science	Object-Oriented Programming (OOP), Algorithms, Problem Solving
Networking & Industrial Systems	TCP/IP, Modbus TCP, PLC Integration, Subnetting, LAN/WAN Basics
Tools	Git/GitHub, VS Code, Arduino IDE
Hardware & System Skills	Sensor Integration, Circuit Assembly, Electronic Troubleshooting, System Integration Testing

Work Experience

Computer Vision Intern Mar 2025 – Jan 2026
Mahindra & Mahindra Ltd. *Nashik, India*

- Designed and implemented an end-to-end software system for automated quality inspection, processing 1,4500+ automotive body panels daily with 98.2% uptime, utilizing Python for scalable data processing and defect detection in CD workflows.
- Integrated solutions with industrial PLC systems via Modbus TCP achieving 94% accuracy, enabling real-time data exchange and control for 40+ configurations, showcasing skills in distributed systems, data engineering, and fault-tolerant software design for large-scale data pipelines.

Secretary Mar 2024 – Present
Sun Nexus Solutions *Nashik,India*

- Coordinated and managed a 30-member technical team, overseeing project discussions, task allocation, and collaboration on innovative technology solutions like AI, IoT, and automation softwares.
- Organized and conducted technical seminars such as "KARMASIDDHI", educating and Mentoring students about different technology domains, career paths, and industry opportunities.
- Conducted market research and developed projects for external clients, gaining practical exposure to real-world problem solving and solution development.

Python Full Stack Developer Intern Jan 2026 – Present
Pentagon Space *Hyderabad, India*

- Pursuing a Python Full Stack Development internship to strengthen software development skills after working on industrial AI projects.
- Enhancing proficiency in Core Python, SQL, and web technologies, focusing on building scalable backend logic and database-driven applications.

Key Projects

Industrial Vision System for Metal Fragment Detection

- Engineered a reverification pipeline with retry logic and adaptive thresholding in Python, attaining 94% detection accuracy while differentiating genuine defects from artifacts in manufacturing workflows, using Scikit-learn for machine learning model implementation.
- Integrated bidirectional PLC handshaking protocols for seamless synchronization, aligning triggers within ±50ms of cycles, showcasing skills in multi-tiered systems, real-time software integration, and data engineering for structured data handling.
- Successfully deployed the system in a live manufacturing environment, enabling real-time defect verification and improving inspection reliability in high-speed production workflows.

AI-Based Vision System for Panel Defect Detection

- Developed an AI-based computer vision system for detecting surface defects on industrial panels using a custom-trained YOLOv8 model, achieving 95% detection accuracy in real-world production conditions.
- Trained and optimized the model on Ada Generation GPUs based on the Ada Lovelace architecture, enabling high-performance model training and faster inference for industrial AI applications.
- Implemented real-time defect detection using 4K IP camera feeds and integrated the system with PLC via Modbus TCP protocol for automated monitoring and low-latency operation on a continuous production line.

Awards & Achievements

- Awarded a certificate of project implementation from **Mahindra & Mahindra Ltd.** for deploying an AI-based vision system in an industrial manufacturing environment.
- Participated in the **NASA Space Apps Challenge**, an international hackathon organized by NASA, collaborating with teams to develop innovative technology solutions for real-world challenges.
- Earned a Merit Award in the **IMPULSE Program** for successfully designing and implementing an Automatic Fire Extinguisher Robot.